Karatina University

Department of Computing & Informatics

COM 332 - Advanced Database Systems (Lab Assignment)

Deadline: 25th September 2025

Create a database to capture the following information;

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

a) Specify the following queries in SQL on the above database schema
   i. Retrieve the names of all senior students majoring in 'cs' (computer science).
   ii. Retrieve the names of all courses taught by Professor King in 1998 and 1999.

iii. For each section taught by Professor King, retrieve the course number, semester, year, and number of students who took the section.
iv. Retrieve the name and transcript of each senior student (Class = 5) majoring in CS. A transcript includes course name, course number, credit hours, semester, year, and grade for each course completed by the student.
v. Retrieve the names and major departments of all straight-A students (students who have a grade of A in all their courses).
vi. Retrieve the names and major departments of all students who do not have a grade of A in any of their courses.

b) Write SQL update statements to do the following on the database schema shown above;
i. Insert a new student, <'Johnson', 25,1, 'MATH'>, in the database.
ii. Change the class of student 'Smith' to 2.
iii. Insert a new course, <Knowledge Engineering', 'cs4390', 3, 'Cs'>.
iv. Delete the record for the student whose name is 'Smith' and whose student number is 17.

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Group Members;

1. **…… Reg No.:……**
2. **…….……Reg No.:…….....**
3. **……..Reg No.:…….**

Sr.	SECTION	x/20	SCORE
	Database Schema		
1.	Tables and field descriptions	2	
2.	Primary keys	2	
3.	Relationships	2	
	Queries		
4.	Retrieve the names of all senior students majoring in 'cs' (computer science).	1	
5.	Retrieve the names of all courses taught by Professor King in 1998 and 1999.	1	
6.	For each section taught by Professor King, retrieve the course number, semester, year, and number of students who took the section.	1	
7.	Retrieve the name and transcript of each senior student (Class = 5) majoring in CS. A transcript includes course name, course number, credit hours, semester, year, and grade for each course completed by the student.	1	
8.	Retrieve the names and major departments of all straight-A students (students who have a grade of A in all their courses).	1	
9.	Retrieve the names and major departments of all students who do not have a grade of A in any of their courses.	1	
10.	Insert a new student, <'Johnson', 25,1, 'MATH'>, in the database.	2	
11.	Change the class of student 'Smith' to 2.	2	
12.	Insert a new course, <Knowledge Engineering', 'cs4390', 3, 'Cs'>.	2	
13.	Delete the record for the student whose name is 'Smith' and whose student number is 17.	2	
	Total		